Social Data - Biases, methodological pitfalls, and ethical boundaries

CT0540 - Social network analysis

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Introduction

Regardless of how large or varied social data are, there are also lingering questions about what can be learned from this data about real-world phenomena e.g.,

- How reflective is user behavior within an online medium of the real-world phenomena being studied?
- How generalizable are the findings from one medium to other media?
- How representative are the working datasets of the targeted platform data?
- How robust are our tools to data biases and variations?
- Are errors evenly distributed across different classes of data?

Prototypical Processing Pipeline for Social Data

General challenges (§3)	**Data processing pipeline**					**Ethical issues (§9)**
	Source (§4)	Collect (§5)	Process (§6)	Analyze (§7)	Evaluate (§8)	
· Population bias · Behavioral bias · Content bias · Linking bias · Temp. variations · Redundancy	· Functional biases · Normative biases · External biases · Non-individuals	· Acquisition · Querying · Filtering	· Cleaning · Enrichment · Aggregation	· Qualitative analysis · Descriptive statistics · Inferences · Observational studies	· Metrics · Interpretation · Disclaimers	· Individual autonomy · Beneficence and non-maleficence · Justice

Prototypical Processing Pipeline for Social Data

- **Data acquisition and preparation**. Gather data collections that satisfy certain characteristics and quality constraints (§3). One would typically identify possible sources of data (§4) from where to extract or sample relevant data through various modes (§5).
- **Data processing**. The acquired data is cleaned, organized and prepared for data analysis (§6). The data points (e.g., users, messages, events) are often represented along a set of features and are categorized along various dimensions.

Prototypical Processing Pipeline for Social Data

- **Data analysis**. Various computational and statistical methodologies can be applied. Their selection depends primarily on the specific settings and goals of each problem—such as describe, infer, predict, or extract causal relations—and on the peculiarities of the working datasets (§7).
- **Evaluation and interpretation**. The results are visualized and evaluated to see, e.g., if they prove or disprove a set of relevant hypotheses, and whether can be generalized beyond a given context (§8). Often, researchers would also attempt to provide an interpretation of the results, which will depend on assumptions made about the meaning of each data point (e.g., whether a social link stems from a friendship relation, or a "like" equates to an endorsement).

Data Quality

For social datasets, researchers have often little leverage to control data quality as they gather data from platforms outside their control:

- **Sparsity**. Social data suffers from sparsity – many measures follow a power law distribution, which makes them easy to mine on the "head," (i.e., in relation to frequent elements or phenomena) but difficult on the long-tail (i.e., rare elements or phenomena). This can be exacerbated by platform functionality design (e.g., limiting or not the size of users' posts)

Data Quality

- **Noise**. Social data suffers from noise (e.g., content that is not reliable or credible, content that is incomplete or corrupted, typos, infrequent terms, stop words) [boyd and Crawford 2012; Naveed et al. 2011]. The distinction between what is "noise" and what is "signal" is often unclear, depending in subtle ways also on the question being asked [Salganik 2017]. As Baeza-Yates [2013] points out, often, the problem is not finding more data, but the right data—e.g. if adding more data also increases the level of noise, the quality and reliability of the results may deteriorate.

Data Quality

- **Representativeness**. Sampling means taking a small portion of elements from a larger population, typically to learn something about the larger population. The key question in sampling is whether the sample is representative or not of the larger population. We have several ways in which to express our representativeness objective:
  - Representativeness with respect to some offline population, (e.g., women who play online games)
  - Representativeness with respect to some notion of relevant content (e.g., content related to sports)
  - Representativeness with respect to some behavior (e.g., people expressing happiness)

Data Quality

- **Data bias**. Data bias is a systematic distortion in the data that compromises its representativeness. Data biases are often evaluated by comparing a data sample with multiple reference samples drawn from different sources or contexts, as obtaining an "absolute" data sample is often unfeasible

General Challenges

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Population Biases

Differences in demographics or other user characteristics between a population of users represented in a dataset or platform and a target population.

- Example: the relationship between a studied population (e.g., adults on Twitter declaring to live in the UK) and a target population (e.g., all adults living in the UK) is often unknown
- Population biases typically refer to the extent to which data collected from a specific social platform differs from a specific target population
- The demographic composition of the major social platforms consistently differ both with respect to each other, as well as with respect to the offline or Internet population

Population Biases

- **Different user demographics tend to be drawn to different social platforms.** Studies show gaps in gender representation across platforms, but also along race, ethnicity, and parental educational background
- **Users of different demographics or traits use social platforms mechanisms differently.** Users of different countries tend to use social media platforms differently. Studies assuming a certain usage may misrepresent certain groups of users.

Population Biases

- **Proxies for personal traits or demographic criteria vary in reliability.** A study interested in the opinion of young college graduates about a new law may rely on a proxy population: those reporting on a social platform to be alumni of a given set of universities. This choice can end up being an important source of bias. In the context of predicting users' political orientations, researchers have shown that the choice of the proxy population drastically influences the performance of various prediction models, producing misleading estimates of users' political orientations on Twitter

Behavioral biases

Differences in user behavior across platforms or contexts, or across users represented in different datasets.

- For instance, information seeking and sharing practices are dependent on the characteristics of the medical condition like being critical or not
- Other studies also look at the similarities and differences among various social platforms w.r.t. adoption patterns, user personalities, news spreading, geographical and socioeconomic patterns, shared content, and behaviors

Behavioral biases

- **Interaction biases affect how users interact with each other**. There are differences in how people communicate on social platforms that are influenced by the relationships they share, and by the features of the platform itself. There are differences in how users balance their attention within their social network with some users being more focused than others, which also depends on their demographics, the type of relation they share, and on shared characteristics
- **Content consumption biases affect how users interact with content**. User consumption behavior is correlated with their demographic attributes and other personal characteristics. Given that users tend to consume more content from like-minded people, such consumption biases are linked to the formation of echo chambers

Behavioral biases

- **The nature of users' tasks influences the traces they leave**. Web search behavior varies across semantic domains, and changes as the task at hand becomes more difficult. Many such tasks also involve searching on behalf of others
- **Misreports and self-selection may occur due to behavioral biases.**
  - Studies relying on self-reports on a certain aspect may be biased due to what users chose to report, when they chose to do it, and how they chose to do it.
  - Many users remain silent despite their interest in a given topic (self-censorship, content removal).
  - Users also have a higher propensity to talk about extreme or positive experiences, rather than talk about negative or common experiences.
  - Inaccurate self-reports can also bias social datasets (*response bias*).

Content Production Biases

> *Behavioral biases that are expressed as lexical, syntactic, semantic, and structural differences in the contents generated by users.*

This bias affects several popular tasks such as user classification, trending topics or content filtering:

- **The use of language(s) varies across and within countries and populations.** Example: By mapping the use of languages across countries, researchers observed seasonal variations in the linguistic composition of each country, as well as between geographical areas at different granularity scales, even at the level of city neighborhoods.

Content Production Biases

- **Contextual factors impact how users talk.** The use of language is even shaped by the relations among users. The temporal orientation of messages may be swayed by factors like openness to experience, number of friends, satisfaction with life, or depression.
- **Different populations have different propensities to talk about certain topics.** Example: African-Americans are more likely to use on Twitter the #BlackLivesMatter hashtag
- **Content from popular or "expert" users differs from regular users' content.** Example: Twitter "expert" users tend to create content mainly on their topic of expertise. Focusing the sampling of content on "expert" users biases the resulting sample towards more trustworthy and high-quality content.

Linking biases

Behavioral biases that are expressed as differences in the attributes of networks obtained from user connections, interactions or activity.

- Linking biases are important for the study of social networks structure, social influence, and information diffusion phenomena:
  - **Network attributes influence users' behavior and perceptions, and vice versa.** Differences in behavior (as indicated by language models and pronoun usage) were found to correlate with users' follower count. Also age-specific graph distances ("degrees of separation") were observed, with younger people being better connected than older generations. Homophily can systematically bias the perceptions of networked users, resulting in under- or over- estimations of the prevalence of certain attributes within a population

Linking biases

Behavior-based and connection-based social links are different. Different graph construction methods can lead to different structural graph properties in the various kinds of networks that can be constructed from social data.

- Exploring the differences between social networks based on explicit connections (users declaring that they are "friends") vs. implicit connections (users interacting repeatedly with each other) showed that the network constructed based on explicit links among users was significantly denser than the one based on user interactions
- In the interaction-defined network, nodes tends to have a smaller, bounded degree.

Linking biases

- Online social networks formation also depends on factors external to the social platforms (e.g., geography). Users visiting the same places being also more likely to create social links online.
- The type and the dynamics of the relationships users share offline influence their propensity to create social ties in online environments and their online interaction patterns.

Temporal Variations

Differences in populations or behaviors over time.

- Temporal variations may affect the ability to understand the generalizability of observations over time (e.g., what factors vary and how they confound with the current patterns in the data)
- Salganik points to three types of variations:
  - Who is using the system
  - How the system is used
  - In the system

Temporal Variations

How one aggregates and truncates datasets along the temporal axis impacts what types of patterns we can observe and what types of research questions we can answer:

- **Populations, behaviors, and systems change over time.** Examples: users' demographic composition and the way in which the platforms are used may change over time, types of interaction (e.g. hashtags)
- **Seasonal and periodic phenomena.** Example: temporal contexts (day vs. night, or weekday vs. weekend) change the shapes of the neighborhoods introduction of a new feature

Temporal Variations

- **Sudden-onset phenomena affect populations, behaviors, and platforms.** Example: suddenly emerging patterns in the data due to external events (e.g., earthquake) or platform changes
- **The time granularity can be too fine-grained to observe long-term phenomena.**(e.g., disasters may have longer term effects than sport events). The time granularity can also be too coarse-grained to observe short-lived phenomena (e.g. for tracking how some experience unfolds for a user)

Temporal Variations

- **Datasets decay and lose utility over time.** This effect has been linked to the deletion of content, as well as to the restrictive nature of the social platforms APIs' terms of service. Cases: The message was explicitly deleted by the user; the user switched their account to "protected" or private; the user's account was suspended; or the user deactivated their entire account.

Redundancy

Single data items that appear in the data in multiple copies, which can be equal to each other (duplicates), or approximately equal to each other (near duplicates).

- It may distort the quantification of phenomena in the data
- There are several sources of content redundancy including the same entity posting from multiple accounts or on multiple platforms (e.g., spam), multiple users posting from the same account (e.g., organization accounts), or multiple accounts posting the same content
- Redundancy can also be a signal by itself! (e.g., if a message is reposted many times)

Issues at the Data Source

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Functional Biases

Biases that are a result of platform-specific mechanisms or affordances, that is, the possible actions within each system or environment.

- Each platform carries its own suite of affordances
  - Actions that can be performed or are encouraged
  - Actions that are not supported or are hard to perform
- Each platform uses dynamic, proprietary, often undocumented platform-specific algorithms to organize and promote content/users

Functional Biases

Platform-specific design and features shape user behavior. Examples include observations of how introducing a new feature, or changing an existing feature on a platform, impacts usage patterns.

- Facebook observed that decreasing the size of the "reply" window to a posting, resulted in users sending shorter replies, faster, and more frequently.
- Also on Facebook, the introduction of the "People You May Know" feature elicited a significant increase in the creation of friendship ties.
- Although distinct social media platforms provide a similar coverage of major events, there are clear differences in the content shared on these platforms.

Functional Biases

- **Algorithms used for organizing and ranking content influence user behavior.** The engagement of users with content, and with other users, is affected by what information is provided to them, when it is provided, and how it is provided.
  - Facebook's algorithmically constructed news feed may influence the ideological diversity of content that users are exposed to
- **The way in which contents are presented influences user behavior.** How different aspects of a data item are organized and emphasized, or the way in which various data items are represented, also impacts how users interact with it and interpret it across platforms. Example: the way emojis are displayed across different smartphone platforms.

Normative Biases

Biases that are a result of written norms or expectations about unwritten norms describing acceptable patterns of behavior on a given platform.

- Each platform is also characterized by certain behavioral norms (what constitutes an acceptable use of each platform)
- **Norms vary across platforms, communities, and contexts. Norms are shaped by user communities.** Users may exhibit different behavior on different platforms: (e.g. family photos on Facebook, not on LinkedIn, trolls and negative feedbacks)

Normative Biases

- **The awareness of being observed by others impacts user behavior.** (*Hawthorne Effect*). Users are more likely to share unpopular opinions/sensitive and personal information in private, anonymous spaces – amplified by privacy concerns (positivity bias)
- **Social conformity and "herding" happen in social platforms.**
  - Example: Prior ratings and reviews, comment length and content
  - Also, tendency to conform with reference points (e.g., influence of user names on perception of quality)

External Biases

Biases resulting from factors outside the social platform, including considerations of socioeconomic status, ideological/religious/political leaning, education, so- cial pressure, privacy concerns, topical interests, language, personality, and culture.

- **Cultural elements and social contexts are reflected in social datasets.**
  - The effect of a particular culture is typically shown through studies comparing a platform's usage across countries.
  - The social context of users in a broader sense also plays a role in their behaviors and interactions
- **High-impact events** are reflected on social media.

Non-individual Accounts

Interactions on social platforms that are not produced by individuals, but by accounts representing various types of organizations, or by automated agents.

- Organizations are widely present on social media
  - These accounts may produce large amount of content (over 60% in crisis events)
- Bots and spamming accounts
  - Not all bots are harmful
  - Some are used to post important updates (weather, emergency alerts)
  - Simply identifying and removing them from the analysis may not be enough (their behavior influences that of human accounts)

Issues in Data Collection

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Issues in Data Collection

Biases introduced due to the selection of data sources, or by the way in which data from these sources are acquired and prepared.

- **Data Acquisition**
- **Data Querying**
- **Data Filtering**

Data Acquisition

- Many social platforms discourage data collection by third parties
- Programmatic access to data from a platform comes with limitations
- The platform may not capture all relevant data (production/interaction vs exposure)
- Platforms may not give access to all the data they capture
- Sampling strategies are often opaque

Data Querying

- APIs have limited expressiveness regarding information needs
  - The specific information needs of a particular research task might not be directly expressible within a particular endpoint
- An information need may be operationalized to an API in different ways
  - There may be more than one formulation to convey an information need
  - e.g., locating social media messages using either their geo-tags or user self-declared geo-location
- The choice of keywords/hashtags shapes the resulting datasets
- Working with the wrong list of keywords might cause relevant data to be missed!

Data Filtering

- Removal of irrelevant portions of the data
- **Outliers are sometimes relevant for data analysis.** Example: Filter out inactive and/or unnaturally active accounts or users from a dataset
- **Text filtering operations may bound certain analyses.** Stopwords may embed useful signals about the valence of the text or the users' personality and psychology

Issues in Data Processing

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Issues in Data Processing

Biases introduced by data processing operations such as cleaning, enrichment, and aggregation.

- **Data Cleaning**: Detecting and correcting errors and inconsistencies in the data (not to be confused with filtering)
- **Data Enrichment**: Adding annotations to data items to be used during the analysis obtained through
  - **Human classification**: (e.g., crowdsourcing, surveys) quality depends on many factors: unreliable annotators, poor guidelines, poor category design, insufficient information to make an assessment
  - **Automatic classification**: Machine learning, NLP classifiers

Issues in Data Processing

- **Data Aggregation**: Structuring, organizing, representing or transforming data. Such transformations can either hide or give prominence to distinct, even divergent, patterns (e.g., aggregating by user may give equal weight to each user's interests, while aggregating by topic may give more weight to content from highly active users)

Methodological Issues

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Methodological Issues

- Prior work raised concerns that the research agenda might be driven by access to data, tools or ease of analysis
- **Using data as a source of hypotheses rather than a tool to test them** or tailoring the research agenda based on data availability, which can result in bias in the type of questions being asked
- Methods:
  - **Qualitative Analyses**
  - **Descriptive Statistics**
  - **Inferences and Predictions**
  - **Observational Studies**

Qualitative Analyses

- Tend to compromise the generalizability for details, particularly due to their limited scope, time or context
- Tend to be sensitive to the interpretation biases of researchers
- They are challenging to scale
- Difficult to draw quantitative evidence from qualitative observations

Descriptive Statistics

- **Social data research often relies on counting entities** (users, links, messages) and computing summaries
  - Simple counts can mislead if it is unclear what is counted and why
  - Example: popular strategies to characterize the emotional state of users rely on counting positive and negative terms, but term frequency is an imprecise measure for how users truly feel
- **Correlation analyses are sensitive to bias**:
  - Example: The extraction of associations among dataset variables with elements of the offline world (e.g., food mentions on social media and obesity rates)
  - The key challenge is how to distill between attributes that merely correlate and those that are causally related

Inferences and Predictions

- **The composition of test and training data samples impacts the results.**
  - Example: using data samples biased towards users whose gender or political identity are easy to discern
  - Performance estimations do not reflect those obtained on balanced or representative samples
- **Distinct target variables, class labels, or data representations may lead to different results.**
  - The data representation or features selected to represent an object (a user, a message) impact the results of inference tasks on those sort of objects.

Challenges with Evaluation and Interpretation of Findings

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Challenges with Evaluation and Interpretation of Findings

Metrics Selection

- Metrics are used to quantify a phenomenon, or the performance of some method or tool.
- But they might be inconclusive, or suffer from reliability and validity issues.
- Two metrics aimed at measuring the same aspect may be inconsistent with one another depending on the context
- Example: The number of posts can be used as a proxy metric for the interest in a topic..
- .. but, while this number may reflect production patterns, it may not reflect how much content on the topic users read

Challenges with Evaluation and Interpretation of Findings

Assessment and Interpretation

- Much research rests upon the assumption that online social traces reflect in real-world phenomena..
- .. but this assumption has been challenged!
- Social media posts should be interpreted as communicative rather than representative
- **The meaning of social traces may change with context**
  - Social links between users can embed different social cues
  - Sharing content can be a sign of endorsement or interest, but users may also share content to disapprove or bully
- **Analyses should go beyond studies confined to a single dataset or method**

Challenges with Evaluation and Interpretation of Findings

Disclaimers and Reproducibility

- Publications of **negative results are scant**
  - Failed studies or negative results are useful for learning about what hypotheses were rejected, or what datasets or methods are not suitable for a given problem
- Disclaimers about the **limitations** of an analysis are **fundamental to good practice**
- If errors or biases cannot be ruled out, researchers must discuss the gaps and limitations in their working datasets, their methods and their assumptions
- Need for data and tools sharing

Other Resources

A selection of reading lists:

- General Issues
- Data Biases and Representativeness
- Methodological Pitfalls and Evaluation
- Ethics of Handling Social Data

Available here

Notes

This presentation is entirely based on Olteanu, A., Castillo, C., Diaz, F., & Kıcıman, E. (2019). Social data: Biases, methodological pitfalls, and ethical boundaries. Frontiers in Big Data, 2, 13.